

Chapter 7

Profit Maximisation and the Competitive Firm

“The purpose of a business is to create a customer.”

— Peter Drucker, *The Practice of Management*, 1954

Before You Begin: Quick Self-Assessment

Test yourself on these prerequisites before reading. Answers are in italics.

1. **What is the relationship between marginal cost (MC) and average cost (AC)?**
When $MC < AC$, average cost is falling; when $MC > AC$, average cost is rising; $MC = AC$ at the minimum of AC.
2. **How do you define economic profit?**
Economic profit = Total Revenue – Total Cost, where total cost includes both explicit (accounting) costs and implicit opportunity costs.
3. **What does it mean for a firm to be a price taker in a perfectly competitive market?**
A price-taking firm is so small relative to the market that it cannot influence the market price; it accepts the going price as given and decides only how much to produce.
4. **What is the short-run shutdown condition, intuitively?**
A firm should shut down when its revenue cannot cover even its variable (avoidable) costs, because every unit produced would then deepen the loss beyond the unavoidable fixed-cost loss.
5. **What is the key difference between the short run and the long run for a firm?**
In the short run, at least one input (typically capital) is fixed and the number of firms in the market is fixed. In the long run, all inputs are variable and firms can freely enter or exit the industry.

Connecting Week 6 to Week 7: Costs Meet Revenue

Week 6 Delivered

Cost curves: SMC, SAC, SAVC, AFC. Cost minimisation: the cheapest way to produce any given output. The LRAC envelope. Short-run vs. long-run cost.

Question left open: Which output level should the firm actually choose?

Week 7 Answers It

Add the revenue side: $TR = P \cdot q$, $MR = P$ (price taker). Combine with cost: $\text{profit} = TR - TC$. Maximise: the $P = SMC$ rule tells the firm exactly what to produce. The cost curves you derived last week become the inputs to this week's output decision.

The payoff of the cost framework: every SMC and SAVC value you computed in Week 6 is used this week to find the optimal quantity, profit, and the shutdown price.

Questions This Chapter Will Answer

- Why does a profit-maximising firm produce where $P = MC$ — and what happens if it produces more or less than that?
- A firm is losing money. Should it shut down? What is the exact threshold?
- How does the firm's short-run supply curve emerge from its cost curves?
- Why does a monopolist's marginal revenue lie below its demand curve?
- How does a demand shock ripple through the firm, market, and consumer simultaneously?
- What keeps economic profit at zero in the long run — and is that a bad thing?

Chapter Road Map. We begin by establishing what it means to be a price taker and showing that marginal revenue equals price for a competitive firm. With revenue pinned down, we derive the profit-maximising condition ($MR = MC$, which becomes $P = SMC$ for a price taker) and learn to read the profit or loss rectangle from a cost-curve diagram. We then develop the shutdown rule — the precise price threshold below which a loss-making firm is better off producing nothing — and use it to derive the firm's short-run supply curve as the SMC curve above $\min(\text{SAVC})$. After a detour showing how marginal revenue falls below price when demand slopes downward (a preview of monopoly in Chapter 8), we aggregate firm supply curves to obtain market supply and trace demand and cost shocks through the three-panel diagram. The chapter closes with the long-run zero-profit equilibrium, the efficiency benchmark against which all market structures are judged.

Learning Objectives and Topics

Each objective states what you will be able to *do*.

LO1 Explain what it means to be a price taker; derive $MR = P$ for a competitive firm.

§7.1–7.3 Price Takers & Revenue

LO2 Apply the $MR = MC$ rule to find the profit-maximising output; verify with the second-order condition.

§7.4 Profit Maximisation

LO3 Compute the profit rectangle $\pi = q^*[P^* - SAC(q^*)]$; classify profit as positive, zero, or negative.

§7.5 Short-Run Profit Maximisation

LO4 Apply the shutdown rule $P \geq \min(SAVC)$; explain why fixed costs are irrelevant to the short-

run decision.

§7.6 Shutdown Decision

LO5 Derive the firm's short-run supply curve from SMC and $\min(SAVC)$.

§7.7 Short-Run Supply Curve

LO6 Derive $MR = a - 2bq$ for a firm facing a downward-sloping demand; explain why $MR < P$.

§7.8 MR Under Downward Demand

LO7 Aggregate firm supply curves into market supply; find short-run equilibrium; trace demand and supply shifts through the three-panel diagram.

§7.9–7.11 Market Equilibrium

Notation Used in This Chapter.

P^*	market (equilibrium) price	TR	total revenue = P^*q
q	firm output	MR	marginal revenue = dTR/dq
Q	market (aggregate) output	MC	marginal cost = dTC/dq
π	economic profit = $TR - TC$	SMC	short-run marginal cost
P_{SD}	shutdown price = $\min(\text{SAVC})$	SAC	short-run average cost
$q^s(P)$	firm's supply function	SAVC	short-run average variable cost
$Q^s(P)$	market supply = $\sum q_i^s$	TFC, TVC	fixed and variable costs

7.1. Introduction: The Competitive Firm

A wheat farmer wakes up to find the world price of wheat is \$200 per tonne. She didn't set that price — no single farmer can. She simply decides how many tonnes to grow. How should she make that decision?

Week 6 Cost Concepts → Week 7 Revenue Concepts

Week 6: Cost Side

SMC(q): slope of STC
 SAC(q): cost per unit (total)
 SAVC(q): cost per unit (variable)
 Cost minimisation: $MRTS = w/v$
 Shape: U-shaped cost curves

Week 7: Revenue Side

TR(q) = $P^* \cdot q$ (price taker)
 MR = P^* (constant)
 $\pi = TR - TC$
 Profit max: $P^* = SMC$
 Shape: profit rectangle

The $P = SMC$ rule this week is the payoff of the entire cost-curve machinery you built in Week 6.

This chapter focuses on one foundational objective: firms seek to *maximise profit*. We begin with perfect competition — the simplest and most instructive market structure — before extending to firms with market power.

Why This Matters. Almost every consumer goods market — commodity grains, gasoline, standardised components — has features close to perfect competition. The tools here explain why agricultural prices fluctuate sharply with weather, why commodity producers respond predictably to cost changes, and why entry and exit keep long-run profits close to zero in competitive industries.

The firm's decision problem has two sides: costs (Week 6) and revenues. We start with the revenue side.

7.2. Price Takers

A single wheat farmer produces 0.001% of world wheat supply. If she doubles her output, does the world price of wheat change? This question defines what it means to be a price taker.

Price Taker: A firm that accepts the market price as given, having no ability to influence it through its own production decisions.

Perfect Competition requires: (1) many small firms each producing a negligible share of market output; (2) identical (homogeneous) products; (3) free entry and exit; (4) perfect information.

In words: A price taker faces a horizontal demand curve at P^ . It can sell any quantity it chooses at that price — but nothing above it.*

Price Taker $\neq$ Indifferent to Output. Being a price taker means the firm cannot influence P^* — not that it is indifferent about how much to produce. The farmer still faces a cost structure that makes producing too little or too much suboptimal. She maximises profit subject to the going price.

Horizontal firm demand at P^ does not contradict the downward-sloping market demand. The market curve shows all buyers; the firm's curve shows what one tiny seller can do.*

Check Your Understanding

1. A competitive market for milk has 500 identical dairy farms. The market demand slopes downward. Why does each farm face a *horizontal* demand curve at price P^* ?
2. One farm tries to charge \$3/litre while all others charge \$2.50. What happens to that farm's sales?
3. True or false: "A price-taking firm has no incentive to reduce its costs, since it cannot change the price anyway."

The price-taking assumption defines the firm's revenue function. Let us now derive revenue concepts explicitly.

7.3. Revenue Concepts for a Price-Taking Firm

At a fixed market price P^ , what happens to total revenue when the firm sells one more unit? The answer — that marginal revenue equals price — is the foundation of the competitive firm's supply decision.*

7.3.1. Total Revenue and Marginal Revenue

Total Revenue:

$$TR(q) = P^* \cdot q \quad (7.1)$$

Marginal Revenue:

$$MR = \frac{\Delta TR}{\Delta q} = P^* \quad (7.2)$$

For a price taker, $MR = P^*$ at every output level. TR is linear in q with slope P^* .

In words: Each extra unit sells at the same price P^ , adding exactly P^* to revenue. MR equals price — always, for a price taker.*

Why $MR = P^*$: Selling an additional unit does not require reducing the price on previously sold units (the firm takes price as given). So each extra unit adds exactly P^* to revenue.

$MR = P^$: the most important special case of the MR formula. Contrast with Week 8 where $MR < P$.*

Example: Bakery at $P^* = \$8$.

At 40 loaves: $TR = 40 \times \$8 = \320 .

At 41 loaves: $TR = 41 \times \$8 = \328 .

$MR = \$328 - \$320 = \$8 = P^*$. ✓

This holds at every quantity: $MR = P^* = \$8$ regardless of whether the bakery sells 1 or 1,000 loaves.

Check Your Understanding

1. For a price taker at $P^* = \$12$: compute TR at $q = 5$, $q = 6$, and $q = 7$. Verify $MR = \$12$ between each pair.
2. A firm facing a downward-sloping demand must lower price to sell more. How does this make its MR differ from P ? (Preview of §7.8.)
3. True or false: “For a price-taking firm, TR is a straight line through the origin.”

We now have the revenue side. Combining revenue with the cost curves from Week 6 gives the profit function — and the rule for maximising it.

7.4. Profit Maximisation

The firm earns \$7 per unit sold. It costs \$4 to produce the next unit. Should it produce that unit? What if it costs \$9? The marginal rule answers both questions with one principle.

7.4.1. The Profit Function

Economic Profit:

$$\pi(q) = TR(q) - TC(q) = P^*q - TC(q) \quad (7.3)$$

The firm chooses $q \geq 0$ to maximise $\pi(q)$.

In words: Profit is the vertical gap between the TR line and the TC curve. The firm’s task is to find where that gap is widest.

7.4.2. The Marginal Rule: $MR = MC$

Intuition Before the Algebra. Adding one more unit has two effects on profit:

- Revenue rises by MR.
- Cost rises by MC.

If $MR > MC$: the extra unit adds to profit — keep expanding.

If $MR < MC$: the extra unit destroys profit — cut back.

Optimum: stop where $MR = MC$.

Formally, differentiate $\pi(q)$ and set to zero:

$$\frac{d\pi}{dq} = \underbrace{\frac{dTR}{dq}}_{MR} - \underbrace{\frac{dTC}{dq}}_{MC} = 0 \implies \boxed{MR = MC} \quad (7.4)$$

In words: Profit is maximised where the revenue from the next unit exactly equals its cost.

Second-order condition: At q^* , the MC curve must cut the MR line *from below*:

$$\left. \frac{dMC}{dq} \right|_{q=q^*} > 0$$

This rules out the possibility that $MR = MC$ at a profit *minimum*. Without the SOC, we cannot distinguish maxima from minima.

MR = MC: the most powerful result in the theory of the firm. Used by every market structure, not just perfect competition.

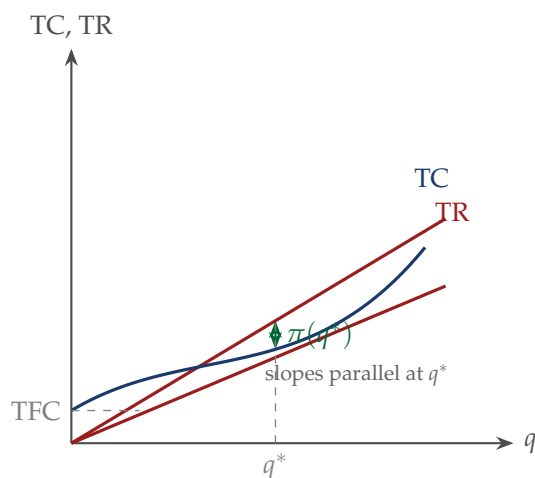


Figure 7.1. TR and TC for a price-taking firm. What to look for: TR is linear (slope = P^*). TC is S-shaped with positive y -intercept TFC. Profit $\pi = TR - TC$ equals the vertical gap. The gap is widest at q^* where the two curves run parallel — i.e., where $MR = MC$ (slopes equal).

Why This Matters. The $MR = MC$ rule is used in every industry: a supermarket deciding whether to stock one more cereal variety, a publisher setting a print run, an airline adding a flight. All face the same marginal logic — add the unit if incremental revenue exceeds incremental cost. This principle also protects against the sunk cost fallacy: only forward-looking marginal costs and revenues matter.

Check Your Understanding

1. At q_0 : $MC = \$4$, $MR = P^* = \$7$. Is the firm at its optimum? Which direction should it adjust? By how much does each unit increase profit?
2. At q_0 : $MC = \$9$, $P^* = \$7$. Same questions.
3. Why is the second-order condition necessary? Give an example of a q where $MR = MC$ but profit is *minimised*.

For a price-taking firm, $MR = P^*$ everywhere. Substituting into the profit-max rule gives the cleanest possible condition: $P^* = SMC$.

7.5. Short-Run Profit Maximisation for a Price-Taking Firm

The market price is $P^* = \$8$. The firm's SMC curve passes through \$8 at two different output levels. How does the firm choose between them — and what does the resulting profit look like on the diagram?

7.5.1. The Optimality Condition: $P^* = SMC$

Because $MR = P^*$ for a price taker, the marginal rule simplifies to:

$$P^* = MR = SMC \quad (7.5)$$

The firm reads off the output where the horizontal price line P^* intersects the upward-sloping SMC curve (second-order condition: the intersection on the *rising* portion of SMC).

In words: The firm asks: "At what output does my marginal cost equal the going market price?" That is the profit-maximising quantity.

Stone Quarry Example. A quarry sells crushed limestone at a regulated rate of \$18/tonne. Its manager finds: $SMC = \$12$ at 500 tonnes, $SMC = \$20$ at 600 tonnes. The profit-maximising output lies between these levels — at approximately 570 tonnes where $SMC = \$18$.

The cost curves from Week 6 are the direct inputs to this calculation.

$P^* = SMC$: one of the most used diagrams in microeconomics. Know it cold.

7.5.2. Measuring Short-Run Profit: The Profit Rectangle

At the optimum q^* , profit equals the area of a rectangle:

$$\pi(q^*) = TR - TC = q^*[P^* - SAC(q^*)] \quad (7.6)$$

In words: Profit per unit = $P^* - SAC(q^*)$. Multiply by the number of units q^* to get total profit.

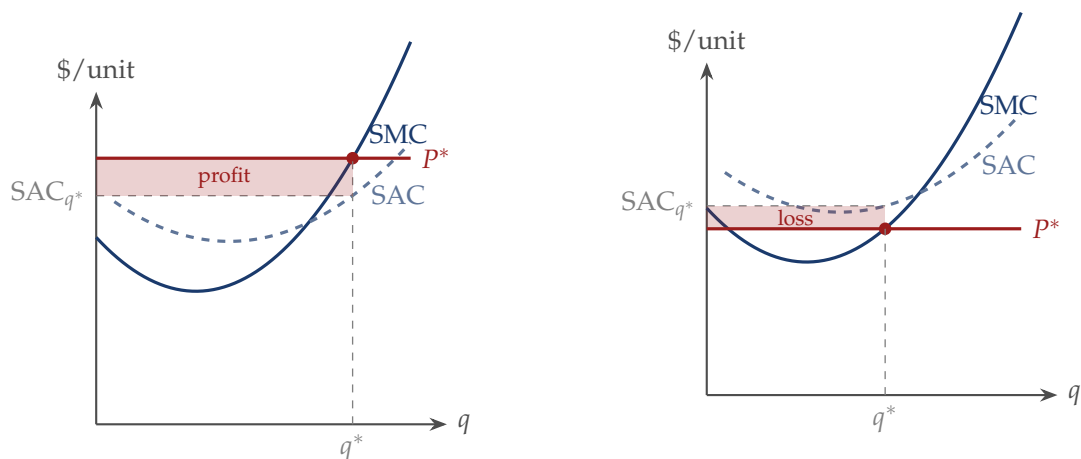
Three cases:

"Which Price Regime Am I In?" — The Decision Guide

Price condition	Profit sign	What to do	Rectangle
$P^* < \min(SAVC)$	$\pi < -TFC$	Shut down ($q = 0$)	No rectangle
$\min(SAVC) \leq P^* < \min(SAC)$	$-TFC < \pi < 0$	Operate at a loss	Height < 0
$P^* = \min(SAC)$	$\pi = 0$	Break-even	Height = 0
$P^* > \min(SAC)$	$\pi > 0$	Positive profit	Height > 0

In our benchmark: shutdown price = $\min(SAVC) = \$4$; break-even price = $\min(SAC) = \$5.50$.

In plain English:
P below \$4: shut down. P between \$4 and \$5.50: run but lose money. P above \$5.50: profitable.



(a) $P^* = \$8 > SAC(q^*)$: positive profit (shaded).

(b) $P^* = \$5$: $\min(SAVC) < P^* < \min(SAC)$, so operate at a loss.

Figure 7.2. Profit and loss rectangles at optimum q^* . What to look for: Profit = (height of rectangle) $\times q^*$. When $P^* > SAC(q^*)$ [left panel], height is positive — profit. When $P^* < SAC(q^*)$ but $P^* > SAVC(q^*)$ [right panel], height is negative — a loss — but the firm still operates because it covers variable costs.

Worked Example: $TC(q) = 2q^2 + 4q + 18$, $P^* = \$24$.

Step 1 — MC: $MC = 4q + 4$.

Step 2 — Set $P = MC$: $24 = 4q + 4 \Rightarrow q^* = 5$.

Step 3 — Profit: $TR = 24 \times 5 = \$120$. $TC = 50 + 20 + 18 = \$88$. $\pi = \$32$.

Step 4 — Rectangle check: $SAC(5) = TC(5)/5 = 88/5 = \17.60 . $\pi = 5 \times (24 - 17.60) = 5 \times 6.40 = \32 . ✓

Check Your Understanding

- Using the decision guide: benchmark curves have $\min(SAVC) = \$4$, $\min(SAC) = \$5.50$. Classify $P^* = \$3$, $P^* = \$5$, $P^* = \$5.50$, $P^* = \$8$.
- For $TC = 0.5q^2 + 5q + 50$: derive MC and SAC. Find $\min(SAC)$ and the break-even price.
- At the break-even price, show algebraically that $\pi = 0$ using the rectangle formula.

We know when profit is positive, zero, or negative. The next question is sharper: when a firm is losing money, should it stop producing entirely?

7.6. The Short-Run Shutdown Decision

A restaurant is losing \$3,000 per month but its fixed costs (rent, insurance) are \$4,000 per month. Should it close? The shutdown rule gives a precise answer — and it is not “close whenever you lose money.”

Common Mistake: Shut down whenever you make a loss. In the short run, this is wrong. A firm that shuts down still incurs fixed costs. The comparison is not between a positive profit and zero, but between two losses: the loss from *producing* vs. the loss from *not producing*. Shut down only when producing makes losses *worse* than $-TFC$.

7.6.1. The Comparison

If the firm produces q^* :

$$\pi(q^*) = TR(q^*) - TVC(q^*) - TFC$$

If the firm shuts down ($q = 0$): $\pi(0) = -TFC$.

Produce if $\pi(q^*) > -TFC$, which simplifies to:

$$TR(q^*) > TVC(q^*) \implies P^* > AVC(q^*)$$

Short-Run Shutdown Rule:

$$P_{SD} = \min(SAVC) \quad (7.7)$$

- If $P^* \geq \min(SAVC)$: **produce** (even if losing money overall).
- If $P^* < \min(SAVC)$: **shut down** ($q = 0$).

In words: Fixed costs are sunk in the short run. Shut down only if you cannot cover variable costs — because every dollar above variable cost goes toward paying off the unavoidable fixed costs.

Solved Example: Restaurant. Monthly $TFC = \$4,000$, $SAVC = \$12/\text{meal}$ at the optimal 500 meals, $P^* = \$14$.

Check: $P^* = \$14 > SAVC = \$12 \Rightarrow$ **operate**.

$$\pi = (14 - 12) \times 500 - 4,000 = 1,000 - 4,000 = -\$3,000.$$

The restaurant loses \$3,000/month, but this beats the shutdown loss of $-\$4,000$.

If P^* falls to \$10: $P^* = \$10 < SAVC = \$12 \Rightarrow$ **shut down**. Every meal deepens the loss beyond $-\$4,000$.

Shutdown price = $\min(SAVC)$. Below this: every unit produced deepens the loss below $-TFC$.

Application: Airlines and the Shutdown Rule.

Airlines famously operate flights even when they do not cover full costs. In the short run, aircraft leases and gate contracts are fixed costs — they are paid whether or not the plane flies. The relevant question is whether *ticket revenue covers variable costs*: fuel, crew wages per flight, cleaning, and catering.

Numbers: Suppose an airline's variable cost per flight is \$25,000 (fuel + crew). Fixed costs (lease, maintenance contracts) allocated to this route are \$30,000/month. Ticket revenue is \$28,000/flight with 20 flights/month.

- Monthly variable cost: $20 \times \$25,000 = \$500,000$.
- Monthly revenue: $20 \times \$28,000 = \$560,000$.

- Contribution to fixed costs: $\$560,000 - \$500,000 = \$60,000$.
- Monthly accounting loss: $\$60,000 - \$30,000 \times (\text{allocated fixed}) = +\$30,000$ profit after contribution.

Even if the route runs at an *accounting* loss, as long as $P^* > \text{SAVC}$ per flight, the airline should keep flying — because flying recovers more of the unavoidable fixed costs than leaving the gate empty.

This is exactly the shutdown rule. Airlines that misapply it (closing routes whenever accounting profit is negative) destroy value by leaving recoverable fixed costs uncovered.

Check Your Understanding

1. A firm has $\text{TFC} = \$500$, $q = 200$, $\text{SAVC} = \$4$, $P^* = \$3.50$. Profit from operating? Loss from shutting down? Which should it choose?
2. Same firm with $P^* = \$5$. Now what? Show using the shutdown rule.
3. True or false: “If $P^* < \text{SAC}(q^*)$, the firm should shut down.” Explain carefully.

The shutdown analysis tells us when quantity supplied equals zero. For prices above the shutdown threshold, the firm produces along its SMC curve. This gives us the supply function.

7.7. The Individual Firm's Short-Run Supply Curve

At every price above \$4 (the shutdown price), the firm produces where $P = \text{SMC}$. Reading this off gives us the supply function. What does it look like?

Short-Run Supply Curve (Individual Firm):

$$q^s(P) = \begin{cases} 0 & \text{if } P < \min(\text{SAVC}) \\ \{q : P = \text{SMC}(q)\} & \text{if } P \geq \min(\text{SAVC}) \end{cases} \quad (7.8)$$

The supply curve is the **upward-sloping portion of SMC at and above $\min(\text{SAVC})$** .

In words: The supply curve is just the SMC curve above the shutdown price. Diminishing returns make SMC slope up, so the firm produces more as price rises.

Supply Curves Come From Cost Curves. The supply curve is derived directly from Week 6 cost curves. An upward-sloping SMC (reflecting diminishing returns to variable inputs) means the firm supplies more as price rises — the law of supply follows directly from the cost structure.

Try It. A firm has $\text{SMC}(q) = 2q + 4$ and $\text{SAVC}(q) = q + 4$. (a) What is $P_{SD} = \min(\text{SAVC})$? (b) Derive $q^s(P)$ for $P \geq P_{SD}$. (c) How much does the firm supply at $P = \$10$?

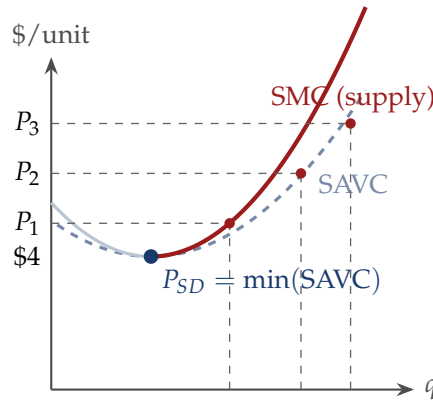


Figure 7.3. The firm's short-run supply curve. *What to look for:* The bold red segment is the supply curve — the SMC curve at and above $\min(\text{SAVC}) = \$4$. At prices below \$4, quantity supplied is zero. As price rises from P_1 to P_2 to P_3 , the firm moves up its SMC, supplying progressively more. The supply curve is upward-sloping because diminishing returns make each additional unit more costly.

Check Your Understanding

1. A firm has $\text{STC} = q^2 + 2q + 9$. Derive SMC and SAVC. Find P_{SD} and $q^s(P)$.
2. A wage increase shifts the SMC curve upward. What happens to the supply curve? Does the firm supply more or less at every price?
3. Why does the supply curve start at $\min(\text{SAVC})$ rather than $\min(\text{SAC})$?

We have derived the supply curve for a price-taking firm. Before building market supply, we explore how MR behaves for a firm not taking price as given — a preview of monopoly in Week 8.

7.8. Marginal Revenue Under a Downward-Sloping Demand Curve

A coffee shop must lower its price from \$5 to \$4.95 to sell one more cup. The 51st cup adds \$4.95 to revenue — but the price cut costs \$0.05 on each of the existing 50 cups. The net marginal revenue is only \$2.45. Why does this happen, and what is the general formula?

7.8.1. Why $\text{MR} < P$ When Demand Slopes Downward

Intuition. For a firm facing downward-sloping demand, selling one more unit requires lowering price *on all units*. The marginal revenue from the extra unit is therefore less than its price, because the price reduction applies to every existing unit.

Two opposing effects when moving from q to $q + 1$:

- **Gain:** revenue from the extra unit sold at the new price P_{new} .
- **Loss:** revenue lost from the price cut on all existing q units.

Net: $\text{MR} = P_{\text{new}} - |\Delta P| \cdot q < P_{\text{new}}$.

In words: MR is less than price because the price cut needed to sell one more unit hurts revenue on all

existing units.

7.8.2. Formula for Linear Demand

Let inverse demand be $P = a - bq$, $a, b > 0$.

$$TR = P \cdot q = (a - bq)q = aq - bq^2.$$

$$MR = \frac{dTR}{dq} = a - 2bq \quad (7.9)$$

MR with Linear Demand: $MR = a - 2bq$ when $P = a - bq$.

The MR curve: same y -intercept as demand but **twice as steep** (slope $-2b$ vs. $-b$). It hits zero at half the quantity where demand hits zero.

In words: Double the slope, same intercept. The MR curve bisects the horizontal distance between the demand curve and the price axis.

Solved Example 4: $P = 10 - q$.

$$TR = (10 - q)q = 10q - q^2; \quad MR = 10 - 2q.$$

At $q = 5$: $P = 5$, $MR = 0$ (TR is maximised).

Numerical verification: $q = 3 \rightarrow 4$

Gain (green): new unit at $P_4 = \$6 \Rightarrow +\6

Loss (red): price drop \$1 on existing 3 units $\Rightarrow -\$3$

$$MR = \$6 - \$3 = \$3 = 10 - 2(3.5) \checkmark$$

(Formula evaluated at midpoint $q = 3.5$.)

*For $P = a - bq$:
 $MR = a - 2bq$. At
 $q = a/(2b)$:
 $MR = 0$ (TR is
 maximised).
 Memorise this.*

Application: $MR < P$ and the Week 8 Preview.

A regional mobile network operator faces a downward-sloping demand. To attract the next subscriber, it must lower its price plan for all existing customers (or offer a cheaper tier). Its marginal revenue from one more subscriber is therefore *less* than the per-subscriber price. This is a key reason monopolists set price *above* marginal cost — they equate $MR = MC$, and since $MR < P$, we get $P > MC$.

This contrasts sharply with a competitive firm, which faces a horizontal demand curve. Its $MR = P$ always. The difference in MR is what makes competitive and monopoly markets produce different outcomes — a central theme of Week 8.

Check Your Understanding

1. For $P = 20 - 3q$: derive TR and MR. At what output is TR maximised? What price holds there?
2. Show algebraically that $MR < P$ for any positive-slope demand curve.

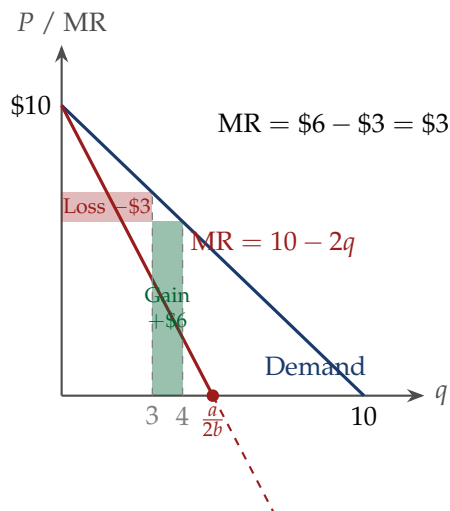


Figure 7.4. MR decomposition for $P = 10 - q$, move $q = 3 \rightarrow 4$. What to look for: Gain (green) = revenue from the extra unit at $P = \$6$; Loss (red) = revenue lost from the \$1 price cut on the existing 3 units. Net MR = $\$6 - \$3 = \$3 = 10 - 2(3.5)$, verified. The MR curve has the same y-intercept as demand but is twice as steep.

3. A price-taking firm's demand is horizontal at P^* . What is MR? How does this compare to the formula $MR = a - 2bq$?

With individual firm supply established, we can now aggregate across all firms to obtain market supply and find the competitive equilibrium.

7.9. Short-Run Market Supply

A city has dozens of bakeries, each with its own upward-sloping supply curve. How do we combine them to get the city's total supply of bread at each price?

Market Supply is the **horizontal sum** of all individual firms' supply curves. If there are n firms and firm i supplies $q_i^s(P)$:

$$Q^s(P) = \sum_{i=1}^n q_i^s(P) \quad (7.10)$$

In words: At each price, add up what every firm supplies. The resulting curve is the market supply curve.

With n identical firms each supplying $q^s(P)$: $Q^s(P) = n \cdot q^s(P)$.

Example: 50 identical firms, $q^s(P) = P - 3$ for $P \geq 3$.

$Q^s(P) = 50(P - 3) = 50P - 150$ for $P \geq 3$.

At $P = \$5.83$: $Q^s = 50(2.83) = 141.5$. Each firm: $q^* = 2.83$. ✓

Check Your Understanding

1. 100 identical firms each have $q^s(P) = (P - 2)/2$ for $P \geq 2$. Write $Q^s(P)$.
2. A new commercial bakery enters with lower costs. How does this affect the market supply curve?
3. True or false: "The market supply curve is steeper than any individual firm's supply curve."

Market supply and market demand together determine the equilibrium price. The three-panel diagram shows how this price is transmitted simultaneously to every firm and every consumer.

7.10. Short-Run Market Equilibrium

The market simultaneously determines a price that balances supply and demand. How does that price propagate to an individual firm's output decision and an individual consumer's purchase decision? The three-panel diagram shows all three at once.

7.10.1. The Three-Panel Framework

Three-Panel Diagram.

1. **Market** (centre): aggregate S and D ; intersection gives P^* and Q^* .
2. **Typical firm** (left): SMC and SAC curves; firm sets $P^* = \text{SMC}$ at q^* .
3. **Typical consumer** (right): individual demand d ; consumer takes P^* as given.

Causality flows outward: the market determines P^* ; the firm and consumer then take P^* as given. No central planner — just prices transmitting information.

7.10.2. Short-Run Response to a Demand Increase

Suppose a health report shifts consumer demand for organic honey outward.

Step 1. Consumer panel: individual demand shifts from d to d' .

Step 2. Market panel: aggregate D shifts to D' ; with supply fixed, P^* rises from P_1 to P_2 and market quantity rises from Q_1 to Q_2 .

Step 3. Firm panel: the horizontal price line shifts from P_1 to P_2 ; each firm moves up its SMC curve, increasing output from q_1 to q_2 . If $P_2 > \text{SAC}(q_2)$, firms earn *positive profit*.

7.10.3. Long-Run Adjustments

In the short run, entry / exit is restricted and capital is fixed. The supply curve is steep: existing firms expand along SMC.

In the long run, two margins open:

- **Input flexibility:** all inputs variable $\Rightarrow$ flatter LMC, LAC.

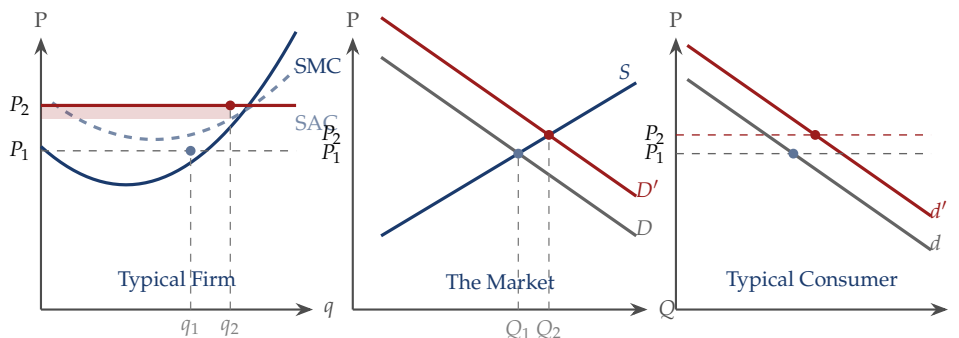


Figure 7.5. Short-run response to a demand increase. What to look for: Consumer demand shifts from d to d' ; market demand shifts from D to D' ; price rises from P_1 to P_2 ; each firm moves up its SMC curve and earns positive profit (shaded rectangle in firm panel). The consumer buys less than before the price rise.

- **Entry/exit:** positive profit attracts entrants $\Rightarrow$ market supply shifts right $\Rightarrow$ price falls until $\pi = 0$.

Long-run equilibrium: $P = LMC = \min(LAC)$, zero economic profit. Firms are healthy and efficient — they earn exactly their opportunity cost.

“Zero economic profit means the firm is in trouble.” This is wrong. Zero economic profit means the firm earns exactly its opportunity cost — owners are compensated at the market rate for all resources, including their own time and capital. A firm in long-run competitive equilibrium is a healthy, efficient firm. It simply earns no *surplus* above normal returns.

Check Your Understanding

1. In a three-panel diagram, trace through the short-run effects of a fall in the price of a complement (reduces demand).
2. Why do positive economic profits attract entry? Why does entry eventually eliminate those profits?
3. Long-run equilibrium has $P = \min(LAC)$. What does this imply about the efficiency of competitive markets?

7.11. Practice: Demand and Supply Shifts

A rise in input prices shifts every firm's SMC upward. A health scare cuts demand for the product. How do you trace both shocks through the three-panel framework?

The three-panel diagram handles any demand or supply shift with the same logic:

Checklist for Demand/Supply Shifts

Shock	Immediate effect	Short-run outcome
Demand rises (sub price $\uparrow$, income $\uparrow$ for normal good)	D shifts right	$P^* \uparrow$, $Q^* \uparrow$, firm earns $\pi > 0$
Demand falls	D shifts left	$P^* \downarrow$, $Q^* \downarrow$, firm earns $\pi < 0$
Input price rises	Firm SMC shifts up	S shifts left, $P^* \uparrow$, $Q^* \downarrow$, $\pi < 0$
Technology improves	Firm SMC shifts down	S shifts right, $P^* \downarrow$, $Q^* \uparrow$, $\pi > 0$

For each scenario: (1) draw the three-panel diagram; (2) identify the direction of change in P^* and Q^* ; (3) determine whether firms earn positive or negative profit in the short run; (4) describe the long-run adjustment (entry/exit until $\pi = 0$).

The content is complete. Before the practice questions, let us review the key results, common mistakes, and reference materials.

Economic Insight. The area *above* the supply curve and *below* the market price is **producer surplus** (PS) — the total gain to the firm from selling at P^* rather than at its reservation cost. PS is the production-side mirror of consumer surplus from Week 4. At the competitive equilibrium, total welfare = CS + PS is maximised. Any price control that pushes the price away from P^* creates a deadweight loss by reducing total surplus — a result developed further in Week 9.

Where Does This Take Us?

- **Week 8 — Monopoly:** A firm with market power faces a downward-sloping demand, so $MR < P$. It sets $MR = MC$, which gives $P > MC$ — a departure from the competitive outcome. The MR formula from Section 7.8 is the foundation of Week 8.
- **Week 9 — Welfare:** $CS + PS = \text{total surplus}$. Competitive equilibrium maximises total surplus. Any intervention (price controls, taxes, subsidies) creates a deadweight loss. The supply curve derived this week is a direct input.
- **Long-run equilibrium:** $P = \min(LAC)$ with zero profit and the optimal number of firms. This is the efficiency benchmark against which all market structures are compared in later chapters.
- **The three-panel framework** is used in every remaining chapter: trade policy (Week 10), externalities, public goods, and factor markets.

Five Common Exam Mistakes — and How to Avoid Them

1. **Using the wrong intersection for q^* .** The SMC curve typically crosses the price line at two points — a minimum and a maximum of the profit function. Always use the intersection on the *upward-sloping* portion of SMC (where $dMC/dq > 0$). The other intersection is a profit *minimum*.
2. **Confusing the shutdown price with the break-even price.** Shutdown price = $\min(SAVC)$. Break-even price = $\min(SAC)$. These are different numbers. The firm shuts down below $\min(SAVC)$, not below $\min(SAC)$. A firm can rationally operate at a loss so long as $P \geq \min(SAVC)$.

3. **Drawing the wrong profit rectangle.** The rectangle's height is $P^* - SAC(q^*)$, not $P^* - SAVC(q^*)$ or $P^* - SMC(q^*)$. Width is q^* . Many students use the wrong cost curve for the bottom of the rectangle.
4. **Treating "zero economic profit" as a bad outcome.** Zero economic profit in the long-run competitive equilibrium means firms earn exactly their opportunity cost on all resources. It is a sign of efficiency, not distress. Accounting profit at this point is typically positive.
5. **Forgetting that market supply is a horizontal sum.** To aggregate supply, add quantities at each price — not prices at each quantity. For n identical firms with supply $q^s(P)$: market supply $Q^s(P) = n \cdot q^s(P)$. If firms differ, sum $q_i^s(P)$ across all types at each given price.

Chapter at a Glance — Week 7: Profit Maximisation

Your one-page revision guide.

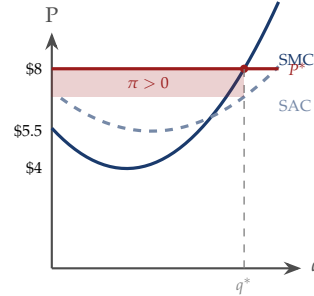
Key Equations

$TR = P^*q$	$MR = P^*$	p. 5
$\pi = TR - TC$	Eq. 7.3	
$MR = MC$	profit max rule	p. 7
$P^* = SMC$	price taker optimum	p. 8
$\pi = q^*[P^* - SAC(q^*)]$	profit rectangle	p. 8
$P_{SD} = \min(SAVC)$	shutdown	p. 10
$q^s(P) = SMC^{-1}(P)$	supply function	p. 11
$Q^s(P) = \sum q_i^s(P)$	market supply	p. 14
$MR = a - 2bq$	linear demand	p. 13

“Which Regime?” Decision Guide

$P < \min(SAVC)$	Shut down
$\min(SAVC) \leq P < \min(SAC)$	Operate (loss < TFC)
$P = \min(SAC)$	Break-even
$P > \min(SAC)$	Positive profit

Firm at Optimum (benchmark)



LO Checklist

- LO1: Price taker; $MR = P^*$ ✓
- LO2: $MR = MC$ rule; SOC ✓
- LO3: Profit rectangle $q^*[P^* - SAC]$ ✓
- LO4: Shutdown rule $P \geq \min(SAVC)$ ✓
- LO5: Supply curve from SMC ✓
- LO6: $MR = a - 2bq$ for linear demand ✓
- LO7: Three-panel; market equilibrium ✓

The thread: Price taker $\rightarrow MR = P^* \rightarrow$ profit max $MR = MC \rightarrow P^* = SMC$ rule $\rightarrow$ profit rectangle $q^*[P^* - SAC] \rightarrow$ shutdown rule $P \geq \min(SAVC) \rightarrow$ supply curve (SMC above shutdown) $\rightarrow$ market supply (horizontal sum) $\rightarrow$ three-panel equilibrium. Week 8 breaks one assumption — price taker — and shows how monopoly pricing differs.

7.12. Key Terms

Price Taker

A firm that accepts the market price as given; faces a horizontal demand curve at P^* . *Used throughout the course.*

Perfect Competition

Market structure: many small price-taking firms, homogeneous product, free entry/exit, perfect information.

Total Revenue (TR)

$P^* \cdot q$; linear in q for a price taker.

Marginal Revenue (MR)

Additional revenue from selling one more unit; $MR = P^*$ for a price taker; $MR = a - 2bq$ for linear inverse demand. *Used throughout the course.*

Marginal Rule

$MR = MC$: the profit-maximising condition for any firm.

$P^* = \text{SMC Rule}$

Profit-maximising condition for a price-taking firm; the special case of $MR = MC$ when $MR = P^*$.

Profit Rectangle

$\pi = q^*[P^* - \text{SAC}(q^*)]$; the geometric representation of profit.

Break-Even Price

$P = \min(\text{SAC})$; the price at which $\pi = 0$.

Shutdown Price

$P_{SD} = \min(\text{SAVC})$; below this, the firm

produces $q = 0$ in the short run. *Used throughout the course.*

Short-Run Supply Curve

The portion of the SMC curve at and above $\min(\text{SAVC})$.

Market Supply

Horizontal sum of all individual firms' supply curves: $Q^s(P) = \sum q_i^s(P)$.

Three-Panel Diagram

Simultaneous depiction of typical firm, market, and typical consumer; causality flows from market to firm/consumer.

Producer Surplus (PS)

Area above the supply curve and below the market price; the mirror of consumer surplus.

Short-Run Equilibrium

$Q^s(P^*) = Q^d(P^*)$ with a fixed number of firms.

Long-Run Equilibrium

$P = \text{LMC} = \min(\text{LAC})$; zero economic profit; entry/exit has ceased.

Supernormal Profit

Positive economic profit; attracts entry in the long run.

Zero Economic Profit

The long-run competitive condition: firms earn exactly their opportunity cost — a healthy, efficient outcome.

Recommended Videos

Four videos selected for clarity and direct relevance to this chapter.

1. Maximising Profit Under Competition (*MRU — Marginal Revolution University*) ~8 min

<https://www.youtube.com/watch?v=0x4f2e5m1Jw>

Explains the $MR=MC$ rule with clear numerical examples and the profit/loss rectangle. Reinforces Sections 7.4–7.5 and ideal before Level 1–2 profit-max questions.

2. Maximising Profit and the Shut-Down Rule (*Jacob Clifford, ACDC Economics*) ~9 min

<https://www.youtube.com/watch?v=wOHbGXOKFFM>

Covers the shutdown rule, AVC vs. ATC, and the three-case profit classification. Directly reinforces Section 7.6 and is recommended before Level 2 shutdown questions.

3. Long-Run Supply in Constant-Cost Markets (*Khan Academy*) ~10 min

https://www.youtube.com/watch?v=Hzw6_vUPvgg

Derives the long-run supply curve, traces entry/exit dynamics, and explains the zero-profit long-run equilibrium. Reinforces Sections 7.9–7.11.

4. Types of Competition and Marginal Revenue (*Khan Academy*) ~8 min

<https://www.youtube.com/watch?v=Jj4hQK-GuXo>

Derives the MR formula for linear demand, explains the gain/loss decomposition, and contrasts MR under competition versus market power. Reinforces Section 7.8 and previews Chapter 8 monopoly.

7.13. Practice Questions

How to use these questions. Difficulty: ★ easy ★★ intermediate ★★★ challenging. Attempt every question before consulting the solutions. Unless stated otherwise, all firms are price takers.

Level 1: Conceptual and Short-Answer

- ★ **[Price Taker — Concept]** Explain in your own words why an individual wheat farmer is a price taker. Does this mean the farmer is indifferent between producing zero and producing a million tonnes? What feature of the market structure ensures no single farmer can influence the price?
- ★ **[MR = Price]** Show algebraically that $MR = P^*$ for a price-taking firm. Explain why this equality would *not* hold for a firm facing a downward-sloping demand curve.
- ★ **[The Marginal Rule — Intuition]** A price-taking firm is currently producing q_0 units.
 - (a) $MC(q_0) = \$4$, $P^* = \$7$. Is the firm at its profit maximum? In which direction should it adjust? How much does each additional unit add to profit?
 - (b) $MC(q_0) = \$9$, $P^* = \$7$. Same questions. Explain the symmetric logic.
- ★ **[Shutdown vs. Producing at a Loss]** A firm has $TFC = \$500/\text{week}$. At its current output of 200 units, $SAVC = \$4$ and $P^* = \$3.50$.
 - (a) Profit from operating at 200 units?
 - (b) Loss from shutting down?
 - (c) Which should the firm choose? Use the shutdown rule.
 - (d) If $P^* = \$5$: repeat.
- ★ **[Supply Curve — Verbal]** Describe in words how a price-taking firm's short-run supply curve is derived from its cost curves. Why does the supply curve only include the SMC portion above $\min(SAVC)$? What happens to the supply curve if the price of a variable input rises?
- ★ **[Long-Run Zero Profit]** In a competitive industry with free entry, economic profit is driven to zero in the long run. Explain why. Does zero economic profit mean firms are doing poorly? How does the result depend on free entry?
- ★ **[Fixed Costs and the Shutdown Decision]** Two economists disagree: one says "fixed costs are irrelevant to production decisions"; the other says "fixed costs determine whether a firm stays in the industry." Reconcile these views. Under what time horizon is each statement correct?
- ★ **[Explain to a Friend — Level 1]** *No formulas.* Your friend says: "If a firm is losing money, it should obviously shut down immediately." Write 3–4 plain-English sentences explaining why this is not always correct, using a concrete example.

Level 2: Intermediate Problems

★★ **[Profit Maximisation — Basic]** $TC(q) = 2q^2 + 4q + 18$, $P^* = \$24$.

- (a) Derive $MC(q)$.
- (b) Find q^* using $P = MC$.
- (c) Calculate $\pi(q^*)$.
- (d) Write $\pi(q)$ explicitly and verify q^* by maximising directly.

★★ **[Average Costs and Profit Measurement]** $TC(q) = q^3 - 6q^2 + 15q + 28$.

- (a) Derive TVC, TFC, SAVC, SAC, MC.
- (b) Find $\min(\text{SAVC})$: the shutdown price.
- (c) Find $\min(\text{SAC})$: the break-even price.
- (d) At $P^* = \$12$: find q^* , π . Should the firm operate?

★★ **[Quadratic Cost — Supply Curve]** $STC(q) = q^2 + 2q + 9$.

- (a) Derive SMC, SAVC, SAC.
- (b) Find $P_{SD} = \min(\text{SAVC})$.
- (c) Derive $q^s(P)$.
- (d) Evaluate q^s at $P = 4, 6, 8, 10$.

★★ **[MR with Linear Demand]** $P = 20 - 3q$.

- (a) Derive TR and MR.
- (b) At what output is TR maximised?
- (c) Verify numerically: compute TR at $q = 3, 4, 5$ and confirm MR matches the formula.
- (d) Explain why $MR < P$ at every positive output.
- (e) Sketch demand and MR on a diagram. Label the output where $MR = 0$.

★★ **[Three Cost Scenarios and Profit]** $TC(q) = 0.5q^2 + 5q + 50$.

- (a) Derive MC and SAC.
- (b) Find $\min(\text{SAC})$ and the break-even price.
- (c) $P^* = \$20$: find q^* and π . Positive, zero, or negative profit?
- (d) $P^* = \$15$: same.
- (e) $P^* = \$10$: use the decision guide to determine whether to operate or shut down.

★★ **[Market Supply with Identical Firms]** 50 identical price-taking firms, each with $q^s(P) = P - 3$ for $P \geq 3$. Market demand $Q^d = 200 - 10P$.

- Derive $Q^s(P)$.
- Find equilibrium P^* and Q^* .
- Each firm's output at P^* ?
- Demand rises to $Q^d = 250 - 10P$. New equilibrium? Change in each firm's output?

★★ **[Data-Based Question]** The table shows a price-taking firm's cost and output data.

q	STC	TFC	TVC	SAVC	SMC
0	50	50	0	—	—
2	70				
4	98				
6	134				
8	178				
10	230				

- Fill in TFC, TVC, SAVC, and SMC for all observations (SMC between consecutive rows).
- What is the shutdown price? At what output does the firm break even?
- If $P^* = \$22$: identify the profit-maximising output (where P^* is closest to SMC). Compute π .
- If $P^* = \$12$: should the firm operate? What is its loss from operating vs. shutting down?
- If $P^* = \$8$: should the firm operate? Justify using the shutdown rule.
- ★ *Week 8 preview*: At the profit-maximising output from (c), compute SAC. Is the firm earning a profit, a loss, or breaking even?

★★ **[Explain to a Friend — Level 2]** Your friend says: "I understand the $MR=MC$ rule, but I don't understand why the supply curve only starts at $\min(SAVC)$ and not at $\min(SAC)$. Surely the firm should shut down whenever it makes a loss?" Write a 4–5 sentence explanation using the concepts of fixed costs and variable costs.

Level 3: Advanced Problems

★★★ **[Cubic Cost — Full Analysis]** $TC(q) = \frac{1}{3}q^3 - 2q^2 + 5q + 12$.

- Identify TFC and TVC.
- Derive MC. Find $\min(MC)$.

- (c) Derive SAVC. Find $P_{SD} = \min(\text{SAVC})$.
- (d) At $P^* = \$5$: find candidates for q^* from $P = MC$. Apply the SOC to identify the true maximum. Compute $\pi(q^*)$. Should the firm operate?

★★★ **[Three-Panel Analysis — Demand Shock]** A competitive market for organic honey is initially in short-run equilibrium.

- (a) Draw a fully labelled three-panel diagram.
- (b) A health report shifts individual consumer demand outward. Describe and draw the short-run adjustment in all three panels.
- (c) Are firms earning positive, negative, or zero profit in the new short-run equilibrium? Show on the firm panel using the SAC curve.
- (d) Describe the long-run adjustment. What happens to entry/exit, market supply, and economic profit?

★★★ **[Three-Panel Analysis — Supply Shock]** A competitive market for disposable cups is initially in equilibrium with $\pi = 0$. World petroleum prices (a key input) rise.

- (a) How does rising input cost shift SMC and SAVC for a typical firm?
- (b) Short-run adjustment in all three panels. Change in equilibrium price, quantity, and firm profit?
- (c) In the short run, do firms earn positive profit, negative profit, or zero profit? Explain carefully.
- (d) Long-run adjustment assuming free entry/exit.

★★★ **[Supply Elasticity with General Cost]** $STC = aq^2 + bq + c$, $a, b, c > 0$.

- (a) Derive SMC and SAVC. Find P_{SD} .
- (b) Derive $q^s(P)$ for $P \geq P_{SD}$.
- (c) Show that the price elasticity of supply $\varepsilon_s = P/(P - b)$. How does ε_s behave as $P \rightarrow \infty$? Interpret.
- (d) With $a = 2, b = 4, c = 10$: evaluate q^s , π , and ε_s at $P = \$10$.

★★★ **[Competitive Equilibrium with 100 Firms]** Inverse market demand $P = 50 - 0.5Q$. 100 identical firms, each with $TC(q) = q^2 + 2q + 25$.

- (a) Each firm's MC and $q^s(P)$.
- (b) Market supply $Q^s(P)$.
- (c) Short-run equilibrium P^* and Q^* .
- (d) Each firm's q^* and π .

- (e) Is this a long-run equilibrium? Compute $\min(\text{SAC})$ and compare to P^* .
- (f) If 150 firms enter, find the new equilibrium price. What does this suggest about the long-run supply curve?

★★★ [Comparative Statics with General Cost] $\text{TC}(q) = q^2 + wq + F$ where $w > 0$ is the wage rate and $F > 0$ is fixed cost.

- (a) Derive $q^*(P, w)$ as a function of P and w .
- (b) Derive $\pi^*(P, w, F)$.
- (c) Find $\partial q^* / \partial P$ and $\partial q^* / \partial w$. Interpret signs.
- (d) Find $\partial \pi^* / \partial P$, $\partial \pi^* / \partial w$, $\partial \pi^* / \partial F$. Verify $\partial \pi^* / \partial P = q^*$ (Hotelling's lemma).
- (e) What is the shutdown price as a function of w ? How does it respond to a wage increase?

★★★ [Long-Run Industry Equilibrium with Entry/Exit] $\text{TC}(q) = q^3 - 4q^2 + 8q + 16$. Market demand $Q^d = 240 - 10P$.

- (a) Find $\min(\text{SAC})$ and the long-run equilibrium price P_{LR} .
- (b) At P_{LR} : find each firm's long-run output q_{LR} from $P = \text{MC}$.
- (c) Find total market quantity and equilibrium number of firms n^* .
- (d) Demand shifts to $Q^d = 300 - 10P$: (i) new short-run P using n^* firms; (ii) are profits positive in the short run? (iii) describe the long-run adjustment and new number of firms.
- (e) Sketch the long-run supply curve. Why is it horizontal for a constant-cost industry?

★★★ [Explain to a Friend — Level 3] Your friend, a business student, says: "Competitive markets are bad for firms — they earn zero profit in the long run. Why would anyone want to be in a competitive industry?" Write 5–6 sentences addressing: (a) what zero *economic* profit actually means; (b) why the competitive outcome is efficient from a social perspective; (c) why it is a better long-run benchmark than industries with persistent monopoly profit; and (d) what a firm in a competitive industry should actually focus on to maximise long-run value.

Full solutions follow in the solutions document. Attempt all questions independently before consulting them.